Amazaon Web Services

Amazaon Web Services Cloud

- It is a collection of remote computing services that together makeup a cloud computing platform offered over the internet .
- AWS is located in nine geographical regions.
- Each region is wholly contained within a single country and all of **its data and services** stay within the designated region.

Amazaon Web Services Cloud

It consist of many cloud services which can be tailored as per the business or organizational need. The services are categorized into different groups.

- Compute
- Storage
- Database
- Migration
- Networking
- Developer Tools

Compute Service

1. Amazon Elastic Compute Cloud (EC2) IaaS

It is a web service that provides secure and resizable compute capacity in the cloud.

Advantages of EC2–

- **Elastic web scale computing** – It helps to increase or decrease computing capacity immediately.
- **Flexible cloud hosting services** – It allows to select memory configuration, CPU, instance storage and boot partition size as per the requirement.
- **Reliable** – It offers a highly reliable environment where replacement instances can be rapidly and predictably commissioned.
- **Secure** – It works in conjunction with Amazon VPC to provide security and robust networking functionality for your computing resources.
- **Inexpensive** – It passes on the financial benefits of Amazon's scale. One has to **pay low rate** for the compute capacity as consumed.

Compute Services ..cont

2.AWS batch –

It enables developers, scientist to easily run thousands of batch computing jobs on AWS.

3. AWS Elastic Beanstack (PaaS)-

It is an easy to use service for deploying and scaling web applications and services.

4. AWS Lambda (PaaS)–

One can run code for virtually any type of application or backend service

Storage Service

1.

Amazon Simple Storage Service (Amazon S3)

It is object storage with simple web service interface to store and retrieve data. It's features are

•Simple – Web based console

•Durable- It provides durable infrastructure to store important data which is stored redundantly across multiple facilities

•Scalable – One can store as much data as needed.

•Available – It is designed for 100% availability over a given period and is backed by S3 SLA.

•Secured – It supports data transfer over SSL and automatic encryption of data once uploaded.

•Low cost – Allows to store large amount of data at low cost.

•Integrated – S3 is deeply integrated with other AWS services to build solutions easily

Storage Service

2.Amazon Elastic Block Store (Amazon EBS)

It provides persistent block storage volume for use with Amazon EC2 instances. Such volumes behave like raw, unformatted storage device, allowing users to create a file system. Each Amazon EBS volume is replicated within it's availability zone to protect from component failure. EBS volumes provide consistent and low latency performance needed to run workloads.

3.Amazon Elastic File System (EFS) –

It provides simple scalable file storage to use with Amazon EC2 instances. It provides standard file system interface and file system access semantics to integrate with existing applications and tools.

4.Amazon Glacier –

It is a secure, durable, low cost storage service for data archiving and long term backup

Amazon Database

1. Amazon Aurora (PaaS)- It is **mySQL compatible relational database engine** that combines the speed & availability of high end commercial databases with simplicity of open source databases.

- High performance
- High available & durable
- Highly scalable
- Highly secure

2.Amazon DynamoDB (PaaS)- It is a fast and flexible **NoSQL database service** for applications that need consistent and less latency.

It supports **both document and key-value data models.**

- Consistent performance
- Flexible
- Highly scalable
- Event driven programming

Migration

Amazon Snowball- It is a large scale data transport solution that uses secure appliances to transfer data into and out of AWS. It uses multilayer security designed to protect data .

- Fast performance
- Highly secured
- Cost effective

Networking

AWS Virtual Private Cloud (VPC) –

It lets user provide a logically isolated section of AWS cloud where one can launch AWS resources in a virtual network . One will have complete control over such virtual networking environment. Additionally one can create a hardware virtual private network connection between corporate data center & VPC and leverage the AWS cloud as an extension of corporate data center.

AWS Direct Connect-

It makes easier to establish dedicated connection from premises to AWS. This connection can be partitioned into multiple virtual interfaces. This allows us to use same connection to access private resources as well as public resources.

Developer Tools

AWS CodeBuild- It is fully managed build service which compiles source code, runs test and produces software packages that are ready to deploy. I t scales continuously and process multiple builds concurrently.

Management Tools

AWS CloudWatch- It is a monitoring service for AWS cloud resources and applications run on AWS. It is used to collect and monitor log files, set alarms and automatically react to changes in AWS resources. It can be used to get information about resource utilization, application performance and operational health for smooth running.